# Section 6.4 Finite Dimensional Spaces

## Exercise 6.4.1

In each case, find a basis for $V$ that includes the vector $v$.

**c.** $V = M_{22}$, $v = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

## Exercise 6.4.2

In each case, find a basis for $V$ among the given vectors.

**b.** $V = P_2$, $\{x^2 + 3, x + 2, x^2 - 2x - 1, x^2 + x\}$

### Exercise 6.4.5

In each case use Theorem 6.4.4 to decide if $S$ is a basis of $V$.

**b.** $V = P_3$; $S = \{2x^2, 1 + x, 3, 1 + x + x^2 + x^3\}$

## Section 7.1 Examples and Elementary Properties

### Exercise 7.1.1

Show that each of the following functions is a linear transformation.

**g.** $T : P_n \to \mathbb{R}$; $T(r_0 + r_1 x + \cdots + r_n x^n) = r_n$

**j.** $T : \mathbb{R}^n \to V;\ T(r_1, \cdots, r_n) = r_1e_1 + \cdots + r_ne_n$ where $\{e_1, \ldots, e_n\}$ is a fixed basis of $V$

## Exercise 7.1.4

In each case, find a linear transformation with the given properties and compute $T(v)$.

**a.** $T : \mathbb{R}^2 \to \mathbb{R}^3;\ T(1, 2) = (1, 0, 1),\ T(-1, 0) = (0, 1, 1);\ v = (2, 1)$

### Exercise 7.1.5

If $T : V \to V$ is a linear transformation, find $T(v)$ and $T(w)$ if:

**a.** $T(v + w) = v - 2w$ and $T(2v - w) = 2v$

## Section 7.2 Kernel and Image of a Linear Transformation

### Exercise 7.2.2

In each case, (i) find a basis of $\ker T$, and (ii) find a basis of $\operatorname{im} T$. You may assume that $T$ is linear.

**b.** $T : P_2 \to \mathbb{R}^2;\ T(p(x)) = (p(0), p(1))$

**e.** $T : M_{22} \to M_{22};\ T\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+b & b+c \\ c+d & d+a \end{bmatrix}$

# Section 7.3 Isomorphisms and Composition

## Exercise 7.3.1

Verify that each of the following is an isomorphism (Theorem 7.3.3 is useful).

**a.** $T : \mathbb{R}^3 \to \mathbb{R}^3;\ T(x, y, z) = (x + y, y + z, z + x)$

**h.** $T : M_{mn} \to M_{nm};\ T(A) = A^T$

## Exercise 7.3.4

In each case, compute the action of $ST$ and $TS$, and show that $ST \neq TS$.

**d.** $S : M_{22} \to M_{22}$ with $S\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$; $T : M_{22} \to M_{22}$ with $T\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & a \\ d & b \end{bmatrix}$

## Exercise 7.3.6

Determine whether each of the following transformations $T$ has an inverse and, if so, determine the action of $T^{-1}$.

**a.** $T : \mathbb{R}^3 \to \mathbb{R}^3;\ T(x, y, z) = (x + y, y + z, z + x)$